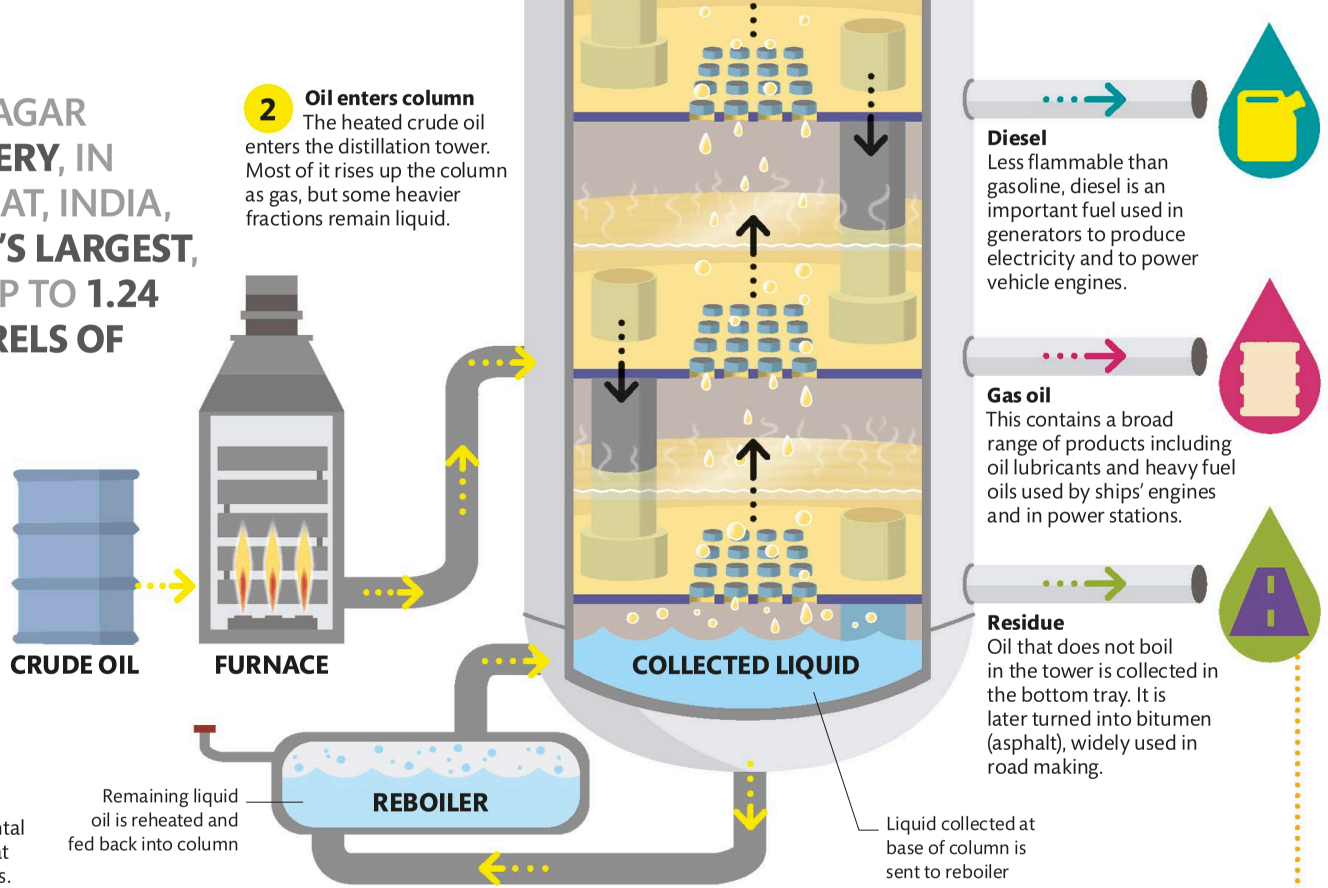
JAMNAGAR **REFINERY**, IN GUJARAT, INDIA, IS THE **WORLD'S LARGEST**, PRODUCING UP TO **1.24 MILLION BARRELS OF OIL A DAY**

1 Desalted crude oil enters furnace
After salts and some other impurities have been removed, crude oil is sent to the distillation furnace. There, it is heated with superheated steam to around 750°F (400°C).

2 Oil enters column
The heated crude oil enters the distillation tower. Most of it rises up the column as gas, but some heavier fractions remain liquid.



Diesel
Less flammable than gasoline, diesel is an important fuel used in generators to produce electricity and to power vehicle engines.

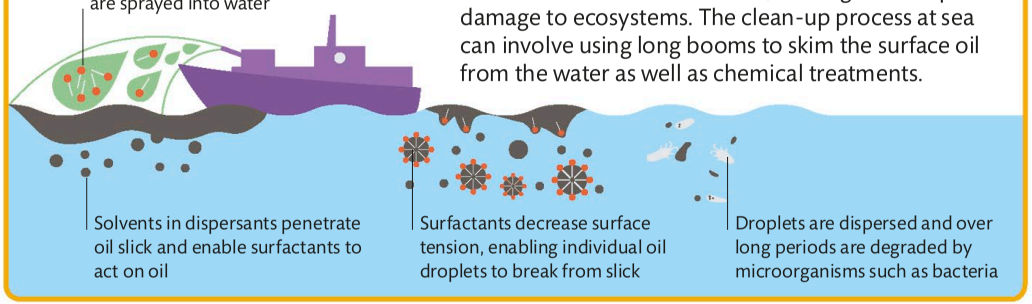
Gas oil
This contains a broad range of products including oil lubricants and heavy fuel oils used by ships' engines and in power stations.

Residue
Oil that does not boil in the tower is collected in the bottom tray. It is later turned into bitumen (asphalt), widely used in road making.

Distillation tower
An oil refinery's distillation tower contains a vertical column divided into horizontal sections containing trays that collect the different fractions.

TREATING OIL SPILLS

Oil tanker accidents and pipeline leaks can release crude oil into the environment, causing catastrophic damage to ecosystems. The clean-up process at sea can involve using long booms to skim the surface oil from the water as well as chemical treatments.



Processing and treatment

Crude oil fractions that have lower boiling points are more flammable and burn with a cleaner flame. As a result, they tend to be in higher demand than heavier fractions. To meet demand, some of the heavier fractions, which are composed of long chains of molecules, are converted into more useful and valuable products through a process called cracking. This often involves breaking down the molecules using heat or a catalyst, such as silicon dioxide or aluminum oxide.